Combined LaTeX Transcriptions

Binder1.pdf Pages 1–20, followed by the first 9 standalone images

This document combines the earlier LaTeX transcriptions into one editable `.tex` file.

Binder1.pdf — Pages 1–20

Page 1

Planetary Rossby Waves

Begin with the assumptions for conserving potential vorticity: inviscid, thin, "rapid" rotation.

$$\frac{D}{Dt}\left(\frac{f+\zeta}{h}\right) = 0$$

The equations describing horizontal motions are:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + fu = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{3}$$

Consider fluctuations with no mean flow (e.g. waves) and small velocities (small Ro), so neglect nonlinear advection terms.

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$$\frac{\partial u}{\partial t} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + fu = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

Substitute for $f = f_0 + \beta y$,

$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

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$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -\frac{1}{\rho}\frac{\partial p}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -\frac{1}{\rho}\frac{\partial p}{\partial y} \tag{2}$$

Replace pressure gradients with surface height anomalies η, that is,

$$\frac{1}{\rho}\frac{\partial p}{\partial x} = g\frac{\partial \eta}{\partial x} \qquad \text{(hydrostatic)}$$

so that

$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -g\frac{\partial \eta}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -g\frac{\partial \eta}{\partial y} \tag{2}$$

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$$\frac{\partial u}{\partial t} - (f_0 + \beta y)v = -g\frac{\partial \eta}{\partial x} \tag{1}$$

$$\frac{\partial v}{\partial t} + (f_0 + \beta y)u = -g\frac{\partial \eta}{\partial y} \tag{2}$$

Recall that the flow is predominantly geostrophic, so the time-dependent and beta terms are small compared to geostrophy.

As the geostrophic terms dominate, use these to express the velocities as

$$u \approx -\frac{g}{f_0}\frac{\partial \eta}{\partial y}, \qquad v \approx \frac{g}{f_0}\frac{\partial \eta}{\partial x}$$

and substitute back into (1,2) for the small terms:

$$-\frac{g}{f_0}\frac{\partial^2 \eta}{\partial y\,\partial t} - f_0 v - \beta y\frac{g}{f_0}\frac{\partial \eta}{\partial x} = -g\frac{\partial \eta}{\partial x} \tag{4}$$

$$\frac{g}{f_0}\frac{\partial^2 \eta}{\partial x\,\partial t} + f_0 u - \beta y\frac{g}{f_0}\frac{\partial \eta}{\partial y} = -g\frac{\partial \eta}{\partial y} \tag{5}$$

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Rearrange (4,5) for v and u:

$$v = \frac{g}{f_0}\frac{\partial \eta}{\partial x} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial y\,\partial t} - \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial x} \tag{6}$$

$$u = -\frac{g}{f_0}\frac{\partial \eta}{\partial y} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial x\,\partial t} + \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial y} \tag{7}$$

The blue terms are ageostrophic. These describe the behaviour of the wave perturbations superimposed on the geostrophic flow.

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Recall (3), the continuity equation,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{3}$$

This can be integrated over a layer depth H,

$$\int_0^H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) dz = H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + w(H) - w(0) = 0$$

$$\frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0 \tag{8}$$

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Substitute u, v from (6,7) into (8):

$$v = \frac{g}{f_0}\frac{\partial \eta}{\partial x} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial y\,\partial t} - \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial x}, \tag{1}$$

$$u = -\frac{g}{f_0}\frac{\partial \eta}{\partial y} - \frac{g}{f_0^2}\frac{\partial^2 \eta}{\partial x\,\partial t} + \beta y \frac{g}{f_0^2}\frac{\partial \eta}{\partial y} \tag{6,7}$$

$$\frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0 \tag{8}$$

$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(\frac{\partial}{\partial x}\left[-f_0\frac{\partial \eta}{\partial y} - \frac{\partial^2 \eta}{\partial x\,\partial t} + \beta y\frac{\partial \eta}{\partial y}\right] + \frac{\partial}{\partial y}\left[f_0\frac{\partial \eta}{\partial x} - \frac{\partial^2 \eta}{\partial y\,\partial t} - \beta y\frac{\partial \eta}{\partial x}\right]\right) = 0$$

$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(-\frac{\partial^3 \eta}{\partial x^2\,\partial t} - \frac{\partial^3 \eta}{\partial y^2\,\partial t} - \beta\frac{\partial \eta}{\partial x}\right) = 0$$

$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(-\frac{\partial}{\partial t}\left[\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}\right] - \beta\frac{\partial \eta}{\partial x}\right) = 0$$

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$$\frac{\partial \eta}{\partial t} + \frac{gH}{f_0^2}\left(-\frac{\partial}{\partial t}\left[\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}\right] - \beta\frac{\partial \eta}{\partial x}\right) = 0$$

Substitute for

$$\nabla^2 \eta = \frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2}$$

and define

$$R = \frac{\sqrt{gH}}{f_0}$$

as the Rossby radius:

$$\frac{\partial \eta}{\partial t} - R^2\frac{\partial}{\partial t}\nabla^2 \eta - R^2\beta\frac{\partial \eta}{\partial x} = 0 \tag{9}$$

This is the Rossby wave equation.

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$$\int_0^H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) dz = \frac{\partial \eta}{\partial t} + H\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) = 0$$

$$\frac{1}{\rho}\frac{\partial p}{\partial x} = g\frac{\partial \eta}{\partial x} \qquad \text{(hydrostatic)}$$

$$R = \frac{\sqrt{gH}}{f_0} \qquad \text{as the Rossby radius}$$

$$\frac{\partial \eta}{\partial t} - R^2\frac{\partial}{\partial t}\nabla^2 \eta - R^2\beta\frac{\partial \eta}{\partial x} = 0$$

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$$\frac{\partial \eta}{\partial t} - R^2 \frac{\partial}{\partial t} \nabla^2 \eta - R^2 \beta \frac{\partial \eta}{\partial x} = 0 \tag{9}$$

This is the Rossby wave equation.

The coefficients of (9) are constant, and we have assumed the perturbation is a wave, so we prescribe the form

$$\eta = \eta_0 \cos(kx - my - \omega t).$$

This allows us to rewrite (9) in terms of ω as a dispersion relation for Rossby waves:

$$\omega = -\beta R^2 \frac{k}{1 + R^2(k^2 + m^2)}$$

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Dispersion relation for planetary Rossby waves

$$\left(\text{recall } R = \sqrt{gH}/f_0\right):$$

$$\omega = -\beta R^2 \frac{k}{1 + R^2(k^2 + m^2)}$$

- For $\beta = 0 \to \omega = 0$: steady, geostrophic flow at constant f (no waves).
- We have assumed $Ro \ll 1$, so solution only valid for $\omega \ll f_0$.
- Consider $k, m \sim 1/L \to \omega \sim \beta R^2 L/(L^2 + R^2)$.
- So for short waves $L < R$: $\omega \sim \beta L$.
- For long waves $L > R$: $\omega \sim \beta R^2/L < \beta L$.
- For $L = R$: ω is maximum at $\omega_{\max} = \beta R/2$.

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Dispersion relation for planetary Rossby waves

$$\left(\text{recall } R = \sqrt{gH}/f_0\right):$$

$$\omega = -\beta R^2 \frac{k}{1 + R^2(k^2 + m^2)}$$

- ZONAL phase speed always negative → westward propagation

$$c_x = \frac{\omega}{k} = -\beta R^2 \frac{1}{1 + R^2(k^2 + m^2)}$$

- MERIDIONAL phase speed either sign

$$c_y = \frac{\omega}{m} = -\beta R^2 \frac{k/m}{1 + R^2(k^2 + m^2)}$$

- Very long waves ($1/k, 1/m, L \gg R$): propagate westward at

$$c = -\beta R^2,$$

  maximum speed allowed.

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$$\frac{D}{Dt}\left(\frac{f+\zeta}{h}\right) = 0$$

North, high f South, low f

Contours of f

Greater f, reduced ζ Smaller f, increased ζ

Greater f, reduced ζ

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Taylor–Proudman Theorem

For steady, $Ro \ll 1$ and uniform density ρ.

Geostrophic vector equation:

$$2\Omega \times \mathbf{u} = -\frac{\nabla p}{\rho}$$

Take the curl of this equation:

$$\nabla \times (2\Omega \times \mathbf{u}) = -\nabla \times \left(\frac{\nabla p}{\rho}\right)$$

RHS: curl of a gradient (∇p) is zero, ρ is uniform.

So:

$$\nabla \times (\Omega \times \mathbf{u}) = 0$$

Expands to:

$$(\Omega \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\Omega + \mathbf{u}(\nabla \cdot \Omega) - \Omega(\nabla \cdot \mathbf{u}) = 0$$

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Taylor–Proudman Theorem

$$(\Omega \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\Omega + \mathbf{u}(\nabla \cdot \Omega) - \Omega(\nabla \cdot \mathbf{u}) = 0$$

Ω is uniform everywhere, and continuity is unaltered by rotation, so:

$$(\Omega \cdot \nabla)\mathbf{u} = 0$$

Choose coordinates such that Ω is in the z-direction,

$$\Omega \frac{\partial \mathbf{u}}{\partial z} = 0 \qquad \rightarrow \qquad \frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = \frac{\partial w}{\partial z} = 0$$

And for a domain with a solid (bottom) boundary where $w = 0$,

$$\frac{\partial u}{\partial z} = \frac{\partial v}{\partial z} = 0, \qquad w = 0$$

Horizontal velocities constant with depth, moves as "Taylor columns".

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Flow on a Rotating Planet – Horizontal Scaling

Euler Equation in Rotating Frame:

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho}$$

x-component of the Euler equation:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + 2\Omega w \cos\theta - 2\Omega v \sin\theta = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$

Assume hydrostatic pressure scale:

$$\Delta P \sim \rho' g H$$

horizontal velocity scale U, timescale $T = L/U$,

$$w \sim \frac{UH}{L}$$

Scales of terms:

$$\frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + 2\Omega\frac{UH}{L} - 2\Omega U \sim \frac{\rho' g H}{\rho L}$$

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Flow on a Rotating Planet – Horizontal Scaling in x

x-component of the Euler equation:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} + 2\Omega w\cos\theta - 2\Omega v\sin\theta = -\frac{1}{\rho}\frac{\partial p}{\partial x}$$

$$\frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + 2\Omega\frac{UH}{L} - 2\Omega U \sim \frac{\rho' g H}{\rho L}$$

$$1 + 1 + 1 + 1 + \frac{2\Omega L}{U}\left(\frac{H}{L} - 1\right) \sim \frac{\rho' g H}{\rho U^2}$$

Rossby Number:

$$Ro = \frac{U}{2\Omega L}$$

$\rightarrow$ ratio of inertial to rotation timescales

$Ro \ll 1$ for "slow" flow, "large" lengthscales and "rapid" rotation rates

$Ro \ll 1 \quad \rightarrow \quad$ Coriolis balances pressure gradient

$$2\Omega v\sin\theta = \frac{1}{\rho}\frac{\partial p}{\partial x}$$

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Flow on a Rotating Planet – Horizontal Scaling in y

y-component of the Euler equation:

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} + 2\Omega u \sin\theta = -\frac{1}{\rho}\frac{\partial p}{\partial y}$$

Scales of terms:

$$\frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + \frac{U^2}{L} + 2\Omega U \sim \frac{\rho' g H}{\rho L}$$

$$1 + 1 + 1 + 1 + \frac{2\Omega L}{U} \sim \frac{\rho' g H}{\rho U^2}$$

Again, Rossby Number:

$$Ro = \frac{U}{2\Omega L}$$

$\rightarrow$ ratio of inertial to rotation timescales

$Ro \ll 1$ for "slow" flow, "large" length scales and "rapid" rotation rates

$Ro \ll 1 \quad \rightarrow \quad$ Coriolis balances pressure gradient

$$-2\Omega u \sin\theta = \frac{1}{\rho}\frac{\partial p}{\partial y}$$

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Navier–Stokes Momentum Equation in Rotating Frame

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho} + \nu \nabla^2 \mathbf{u}$$

Euler Equation in Rotating Frame

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho}$$

$$2\Omega \times \mathbf{u} = (-fv,\, fu,\, 0)$$

$$f = 2\Omega \sin\theta$$

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Flow on a Rotating Planet – Vertical Scaling

Euler Equation in Rotating Frame:

$$\frac{D\mathbf{u}}{Dt} + 2\Omega \times \mathbf{u} = \mathbf{g} - \frac{\nabla p}{\rho}$$

Vertical component of the Euler equation:

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} - 2\Omega u \cos\theta = -g - \frac{1}{\rho}\frac{\partial p}{\partial z}$$

$$H \ll L$$

$$T = \frac{L}{U}$$

$$\frac{U}{L} \sim \frac{W}{H} \qquad \Rightarrow \qquad W \sim \frac{UH}{L}$$

Standalone Images 1–9

Image 1

This is the 1st generation only. We have two more copies of all these rows, identical in every way except for mass. 2nd and 3rd gen matter is unstable!

1st Gen	**2nd Gen**	**3rd Gen**
Electron e	Muon μ	Tau τ
Electron Neutrino ν_e	Muon Neutrino ν_μ	Tau Neutrino ν_τ
Up Quark u	Charm Quark c	Top (Truth) Quark t
Down Quark d	Strange Quark s	Bottom (Beauty) Quark b

$$T_3 \equiv \tau_3/2$$

So, visually it's like:

$$Q = \frac{1}{2}\left(\tau_3 + Y\right)$$

rotate around z-axis rotate around flagpole

Particle	**Electric Charge Q**	**Weak Isospin T_3**	**Hypercharge Y**
Electron e_R	-1	0	-2
Neutrino ν_e	0	$+1/2$	-1
Electron e_L	-1	$-1/2$	-1
Up Quark u_R	$+2/3$	0	$+4/3$
Down Quark d_R	$-1/3$	0	$-2/3$
Up Quark u_L	$+2/3$	$+1/2$	$+1/3$
Down Quark d_L	$-1/3$	$-1/2$	$+1/3$
$W^\pm$ Bosons	± 1	± 1	0
Z Boson	0	0	0
Photon γ	0	0	0
Gluon	0	0	0
Higgs H^+	$+1$	$+1/2$	$+1$
Higgs H^0	0	$-1/2$	$+1$

Leptons Quarks Gauge Bosons Higgs Field

$$SU(2)_L \text{ doublets: } \begin{pmatrix} \nu_e \\ e_L \end{pmatrix}, \quad \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}$$

$$\text{Gell-Mann–Nishijima relation:} \qquad Q = T_3 + \frac{Y}{2}$$

Image 2

$$\Psi = \begin{pmatrix} \psi_r \\ \psi_g \\ \psi_b \end{pmatrix}, \qquad \bar{\Psi} = \begin{pmatrix} \bar{\psi}_r & \bar{\psi}_g & \bar{\psi}_b \end{pmatrix}$$

Image 3

The color triplet field $\Psi \in \mathbb{C}^3 \otimes \mathbb{C}^4 \cong \mathbb{C}^{12}$ gives us a way to write $\mathcal{L}$ more concisely, but now that

we're here... what happens if we require $\mathcal{L}$ to remain unchanged under any transformation

$$\Psi \to \Psi' = U(x)\Psi \qquad \text{where } U(x) \in SU(3)?$$

Image 4

Free quark, non-interacting three-color Dirac lagrangian:

$$\mathcal{L} = \mathcal{L}_r + \mathcal{L}_g + \mathcal{L}_b$$
$$\mathcal{L} = \bar{\psi}_r \left(i\gamma^\mu \partial_\mu - m\right) \psi_r + \bar{\psi}_g \left(i\gamma^\mu \partial_\mu - m\right) \psi_g + \bar{\psi}_b \left(i\gamma^\mu \partial_\mu - m\right) \psi_b$$

$$\mathcal{L} = \bar{\Psi} \left(i\gamma^\mu \partial_\mu - m\right) \Psi$$

Image 5

Free quark, non-interacting three-color Dirac lagrangian:

$$
\begin{aligned}
\mathcal{L} &= \mathcal{L}_r + \mathcal{L}_g + \mathcal{L}_b \\
\mathcal{L} &= \bar{\psi}_r \left(i\gamma^\mu \partial_\mu - m\right) \psi_r \\
&\quad + \bar{\psi}_g \left(i\gamma^\mu \partial_\mu - m\right) \psi_g \\
&\quad + \bar{\psi}_b \left(i\gamma^\mu \partial_\mu - m\right) \psi_b
\end{aligned}
$$

Image 6

$$
\begin{aligned}
\psi_r &= u_s(\vec{p}) \exp[-i\left(Et - \vec{p}\cdot\vec{x}\right)]\,, \\
\psi_g &= u_s(\vec{p}) \exp[-i\left(Et - \vec{p}\cdot\vec{x}\right)]\,, \\
\psi_b &= u_s(\vec{p}) \exp[-i\left(Et - \vec{p}\cdot\vec{x}\right)]\,.
\end{aligned}
$$

Image 7

Baryon	**Quark content**	**Charge**	**Mass** (GeV/c^2)	**Half Life (seconds)**	**Principal decays**	**Spin**
p	uud	$+1$	0.93828	at least $3 \cdot 10^{41}$	–	1/2
n	udd	0	0.93957	880	$pe\bar{\nu}_e$	1/2
Λ	uds	0	1.1156	$2.63 \cdot 10^{-10}$	$p\pi^-$, $n\pi^0$	1/2
Σ^+	uus	$+1$	1.1894	$0.80 \cdot 10^{-10}$	$p\pi^0$, $n\pi^+$	1/2
Σ^0	uds	0	1.1925	$6 \cdot 10^{-20}$	$\Lambda\gamma$	1/2
Σ^-	dds	-1	1.1973	$1.48 \cdot 10^{-10}$	$n\pi^-$	1/2
Ξ^0	uss	0	1.3149	$2.90 \cdot 10^{-10}$	$\Lambda\pi^0$	1/2
Ξ^-	dss	-1	1.3213	$1.64 \cdot 10^{-10}$	$\Lambda\pi^-$	1/2
Λ_c^+	udc	$+1$	2.281	$2 \cdot 10^{-13}$	$pK^-\pi^+$	1/2
Δ	uuu, uud, udd, ddd	$+2$, $+1$, 0, -1	1.232	$0.6 \cdot 10^{-23}$	$N\pi$	3/2
Σ^*	uus, uds, dds	$+1$, 0, -1	1.385	$2 \cdot 10^{-23}$	$\Lambda\pi$, $\Sigma\pi$	3/2
Ξ^*	uss, dss	0, -1	1.533	$7 \cdot 10^{-23}$	$\Xi\pi$	3/2
Ω^-	sss	-1	1.672	$0.82 \cdot 10^{-10}$	ΛK^-, $\Xi^0\pi^-$, $\Xi^-\pi^0$	3/2

Source: *Introduction to Elementary Particles*, David Griffiths, table at the beginning of the book.

Image 8

Vector Meson (Spin-1)	**Quark content**	**Charge**	**Mass** (GeV/c^2)	**Half Life (seconds)**	**Principal decays**
ρ	$u\bar{d}$, $d\bar{u}$, $(u\bar{u}-d\bar{d})/\sqrt{2}$	$+1$, -1, 0	0.770	$0.4 \cdot 10^{-23}$	$\pi\pi$
K^*	$u\bar{s}$, $s\bar{u}$, $d\bar{s}$, $s\bar{d}$	$+1$, -1, 0, 0	0.892	$1 \cdot 10^{-23}$	$K\pi$
ω	$(u\bar{u}+d\bar{d})/\sqrt{2}$	0	0.783	$7 \cdot 10^{-23}$	$\pi^+\pi^-\pi^0$, $\pi^0\gamma$
ϕ	$s\bar{s}$	0	1.020	$20 \cdot 10^{-23}$	K^+K^-, $K^0\bar{K}^0$
J/ψ	$c\bar{c}$	0	3.097	$1 \cdot 10^{-20}$	e^+e^-, $\mu^+\mu^-$, 5π, 7π
D^*	$c\bar{d}$, $d\bar{c}$, $c\bar{u}$, $u\bar{c}$	$+1$, -1, 0, 0	2.010	$> 1 \cdot 10^{-22}$	$D\pi$, $D\gamma$
Υ	$b\bar{b}$	0	9.460	$2 \cdot 10^{-20}$	$\tau^+\tau^-$, $\mu^+\mu^-$, e^+e^-

Image 9

Pseudoscalar Meson (Spin-0)	**Quark content**	**Charge**	**Mass** (GeV/c^2)	**Half Life (seconds)**	**Principal decays**
$\pi^\pm$	$u\bar{d}$, $d\bar{u}$	$+1$, -1	0.139569	$2.60 \cdot 10^{-8}$	$\mu\nu_\mu$
π^0	$(u\bar{u}-d\bar{d})/\sqrt{2}$	0	0.134964	$8.7 \cdot 10^{-17}$	$\gamma\gamma$
$K^\pm$	$u\bar{s}$, $s\bar{u}$	$+1$, -1	0.49367	$1.24 \cdot 10^{-8}$	$\mu\nu_\mu$, $\pi^\pm\pi^0$, $\pi^\pm\pi^\pm\pi^\mp$
K^0, $\bar{K}^0$	$d\bar{s}$, $s\bar{d}$	0, 0	0.49772	$K_S^0 : 0.892 \cdot 10^{-10}$	$\pi^+\pi^-$, $\pi^0\pi^0$
K^0, $\bar{K}^0$	$d\bar{s}$, $s\bar{d}$	0, 0	0.49772	$K_L^0 : 5.18 \cdot 10^{-8}$	$\pi e\nu_e$, $\pi\mu\nu_\mu$, $\pi\pi\pi$
η	$(u\bar{u}+d\bar{d}-2s\bar{s})/\sqrt{6}$	0	0.5488	$7 \cdot 10^{-19}$	$\gamma\gamma$, $\pi^0\pi^0\pi^0$, $\pi^+\pi^-\pi^0$
η'	$(u\bar{u}+d\bar{d}+s\bar{s})/\sqrt{3}$	0	0.9576	$3 \cdot 10^{-21}$	$\eta\pi\pi$, $\rho^0\gamma$
$D^\pm$	$c\bar{d}$, $d\bar{c}$	$+1$, -1	1.869	$9 \cdot 10^{-13}$	$K\pi\pi$
D^0, $\bar{D}^0$	$c\bar{u}$, $u\bar{c}$	0, 0	1.865	$4 \cdot 10^{-13}$	$K\pi\pi$
$D_s^\pm$	$c\bar{s}$, $s\bar{c}$	$+1$, -1	1.971	$3 \cdot 10^{-13}$	not established
$B^\pm$	$u\bar{b}$, $b\bar{u}$	$+1$, -1	5.271	$14 \cdot 10^{-13}$	$D+$?
B^0, $\bar{B}^0$	$d\bar{b}$, $b\bar{d}$	0, 0	5.275	$14 \cdot 10^{-13}$	$D+$?
η_c	$c\bar{c}$	0	2.981	$6 \cdot 10^{-23}$	$KK\pi$, $\eta\pi\pi$, $\eta'\pi\pi$